ALLEN CHANDEV

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EDUCATION

St Hilda's School Ooty 2019 - 2025

ICSE

Karunya Institute of Technology and Sciences 2025 - 2029

B.Tech Computer Science Engineering

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, C/C++, HTML/CSS
Frameworks	React, Vite, Flask, Tailwind CSS, Tkinter
Tools	Ollama, Firebase, Git/GitHub, VS Code

FEATURED PROJECTS

PathX *[React 19, TypeScript, Vite 6, Motion]*

Autonomous Vehicle Simulator

- Engineered a high-fidelity simulator leveraging **Kinematic Bicycle Models** and **Pure Pursuit algorithms** at 60Hz.
- Implemented mathematical foundations using **PDEs** for state change modeling and **Laplace Transforms** for control system stability.
- Designed a premium dark-mode dashboard with real-time telemetry visualization using **D3-driven logic**.

Lumina AI Assistant *[React, TypeScript, Ollama (Llama-3)]*

Privacy-First AI Tutoring Platform

- Architected a privacy-first AI platform by migrating external API dependencies to **locally-hosted inference pipelines (Ollama)**.
- Integrated speech-to-text mapping and dynamic interactive Flashcards directly from LLM inferences.
- Developed a sophisticated React frontend with state-driven conversational UI logic.

Edgelytics *[Python, Flask, Arduino DHT22]*

Edge Computing IoT Framework

- Constructed an edge-computing framework to ingest and manage real-time telemetry from Arduino microcontrollers.
- Implemented a **local anomaly-detection algorithm** to trace impossible environmental data spikes.
- Integrated device metrics through a scalable Python Flask backend to a live dashboard.

Emergency Geo-Router *[Geolocation API, React]*

Fault-Tolerant Redundant Routing System

- Deployed a fault-tolerant routing application to triangulate and transmit location data in offline-critical environments.
- Wrote robust handling algorithms to detect low-accuracy GPS metrics and execute cascading fallbacks.

NextChapter *[Python, Tkinter]*

Personalized Book Recommendation System

- Refactored a desktop-based Python application handling complex graphical frame life-cycles and state management.
- Re-architected the state pipeline for a personalized book recommendation engine using optimized object structures.

PROFILE SUMMARY

B.Tech CS student at Karunya University with a focus on high-fidelity simulation, AI platform architecture, and innovative engineering solutions. Dedicated to building secure, scalable, and impactful technical systems.